Jiaqi Chen

Ph.D. Student, Department of Electrical and Computer Engineering, UW Madison

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Education

University of Wisconsin–Madison Madison, WI, United States

Ph.D. student in Electrical and Computer Engineering (major), Computer Science (minor) *Sep. 2020 – ongoing*

- Supervisor: Prof. Line A. Roald (IEEE member)
- Research interests: data-driven method and its application in power systems, modelling, optimization and control in distribution systems, stochastic optimization
- GPA: 3.94 out of 4

Tsinghua University Beijing, P.R. China

M.S. in Electrical Engineering *Sep. 2017 – May 2020*

- Supervisor: Prof. Wenchuan Wu (IEEE fellow)
- Research interests: data-driven algorithms in power systems, electricity market in distribution networks, optimization, operation and control in unbalanced three-phase active distribution networks
- Master Thesis: "Data-driven Three-phase Power Flow Linearization for Active Distribution Networks and Its Application"
- GPA: 3.67 out of 4

Shandong University Jinan, Shandong Province, P.R. China

B.S. in Electrical Engineering and Automation *Sep. 2013 – June 2017*

- Bachelor Thesis: "A Distributed Reactive Power Optimization and Control Method for Active Distribution Networks"
- GPA: 92.82 out of 100 (Ranking: 1st); honored as "President's Scholarship" in Shandong University (10 out of all students)

Publications

[C1] **J. Chen**, and L. A. Roald, "Real-time assessment of distribution grid security through adaptive smart meter measurements," submitted to *2024 63rd IEEE Conference on Decision and Control (CDC).*

[C2] **J. Chen**, and L. A. Roald, "Topology-Adaptive Piecewise Linearization for Three-Phase Power Flow Calculations in Distribution Grids," *2023 North American Power Symposium (NAPS)*, Asheville, NC, USA, 2023, pp. 1-6.

[J1] **J. Chen** and L. A. Roald, "A data-driven linearization approach to analyze the three-phase unbalance in active distribution systems," *Electric Power Systems Research*, vol. 211, p. 108573, 2022.

[J2] **J. Chen**, W. Wu, and L. A. Roald, "Data-driven piecewise linearization for distribution three-phase stochastic power flow," *IEEE Transactions on Smart Grid*, vol. 13, no. 2, pp. 1035–1048, 2021.

[C3] J. Lindberg, Y. Abdennadher, **J. Chen**, B. C. Lesieutre, and L. Roald, "A guide to reducing carbon emissions through data center geographical load shifting," *Twelfth ACM International Conference on Future Energy Systems*, 2021, pp. 430–436.

[C4] **J. Chen**, W. Li, W. Wu, T. Zhu, Z. Wang, and C. Zhao, "Robust data-driven linearization for distribution three-phase power flow," *2020 IEEE 4th Conference on Energy Internet and Energy System Integration (EI2).* IEEE, 2020, pp. 1527–1532.

[J3] **J. Chen**, Y. Guo, and W. Wu, "Optimal dispatch scheme for DSO and prosumers by implementing three-phase distribution locational marginal prices," *IET Generation, Transmission & Distribution*, vol. 14, no. 11, pp. 2138–2146, 2020.

[C5] T. Xu, W. Wu, **J. Chen**, Q. Lu, and Y. Zhu, "Asynchronous distributed dynamic economic dispatch in active distribution networks," *2019 IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia)*. IEEE, 2019, pp. 318–322.

[C6] Z. Hou, Y. Liang, Y. Zhang, W. Wu, and **J. Chen**, "Security evaluation under N-1 for active distribution networks coordinated with transmission grid," *2019 IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia)*. IEEE, 2019, pp. 710–715.

[C7] Q. Lu, Y. Yang, Y. Zhu, T. Xu, W. Wu, and **J. Chen**, "Distributed economic dispatch for active distribution networks with virtual power plants," *2019 IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia)*. IEEE, 2019, pp. 328–333.

[C8] Q. Lu, **J. Chen**, Y. Zhu, S. Liu, Y. Xu, and K. Wang, "Risk assessment with high distributed generations penetration considering the interaction of transmission and distribution grids," *2018 2nd IEEE Conference on Energy Internet and Energy System Integration (EI2)*. IEEE, 2018, pp. 1–6.

Professional Services

Reviewer: IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Power Engineering Letters, IET Generation, Transmission & Distribution, IEEE Power Energy Society General Meeting, Power Systems Computation Conference, IEEE Conference on Energy Internet and Energy System Integration.

Member: IEEE, IEEE Power Energy Society, PES Women in Power, UW-Madison Women in ECE.

Project and Internship Experience

Graduate Research Assistant | *Electric Power Systems Research Group, UW-Madison* Sep. 2021 – ongoing

- Participate in the project *Smart Meter-Driven Distribution Grid Visibility and Control*, developing data-driven models for distribution systems and associated methods for real-time state estimation and control applications (PSERC T-67)
- Participate in the project reducing carbon emissions through data center geographical load shifting, from Reactive to Proactive Distribution Grid Risk Management, supported by National Science Foundation (NSF)
- Research on data-driven method and its application in power systems, optimization and control in active distribution systems. Develop data-driven piecewise linear power flow model and the linear approximation for three-phase unbalance metric in distribution systems.

Research Assistant | *State Key Lab of Control and Simulation of Power Systems, Tsinghua Univ.* Sep. 2017 – Jul. 2021

- Participate in the project in control strategy and security evaluation considering the interaction of transmission and distribution grids, supported by National Science Foundation of China (NSFC).
- Research on optimal dispatch and pricing scheme based on three-phase distribution locational marginal prices, security evaluation and control strategy for distribution grids coordinated with transmission grids.

M.S. Summer Intern | *Heying Internet technology Co. Ltd.* Jul. 2018 – Aug. 2018

- Data analysis and data mining for consumers' data of sports watches. Participate in developing the software of the sports watch.

Awards & Scholarships

2023 Sixth Workshop on Autonomous Energy Systems Travel Grant, AES Workshop Organizing Committee
2023 ECE Dissertator Travel Award, UW-Madison
2020 Wisconsin Distinguished Graduate Fellowship-Procknow (WDGF), UW-Madison
2020 ECE Graduate Welcome Award, UW-Madison
2017 Outstanding Contribution Award, 2017 IEEE Conference on Energy Internet and Energy System Integration
2017 Outstanding Graduate Award and Excellent Graduation Thesis Award, Shandong University
2016 Outstanding Student, Shandong Province
2015 Second Prize of 2015 International Contest of Innovation
2015 Champion of Help Service Robot Challenge, RoboCup China
2015 President's Scholarship in Shandong University (10 out of all students)
2015 UHV Scholarship, Shandong University (10 out of all students)
2014-2016 China National Scholarship (two times)
2014-2016 First Prize Scholarship (three times), Shandong University

Presentations and Posters

Seventh Workshop on Autonomous Energy Systems, (Golden, CO, USA, Sept., 2024)
Real-time Assessment of Distribution Grid Security through Adaptive Smart Meter Measurements, (presentation)

2024 IEEE PES General Meeting, (Seattle, Wa, USA, Jul., 2024)
Real-time Assessment of Distribution Grid Security through Adaptive Smart Meter Measurements, (poster)

PSERC's 2023 December IAB meeting, (Atlanta, GA, USA, Dec. 2023)
Data-driven Piecewise Linearization for Three-Phase Power Flow in Distribution Grids, (poster)

The 55th North American Power Symposium (NAPS), (Asheville, NC, USA, Oct. 2023)
Topology-Adaptive Piecewise Linearization for Three-Phase Power Flow Calculations in Distribution Grids, (presentation)

Sixth Workshop on Autonomous Energy Systems, (Golden, CO, USA, Sept., 2023)
Data-driven Piecewise Linearization for Three-Phase Power Flow in Distribution Grids, (presentation)

2023 IEEE PES General Meeting, (Orlando, FL, USA, Jul., 2023)
Data-driven Piecewise Linearization for Three-Phase Power Flow in Distribution Grids, (poster)

2023 Grid Science Winter School and Conference, (Santa Fe, NM, USA, Jan., 2023)
Data-driven Piecewise Linearization for Three-Phase Power Flow in Distribution Grids, (poster)

Power Systems Computation Conference (PSCC), (Porto, Portugal, Jul., 2022)
A data-driven linearization approach to analyze the three-phase unbalance in active distribution systems, (presentation)

DTU Summer School 2022, (Copenhagen, Denmark, Jun., 2022)
Data-driven piecewise linearization for distribution three-phase stochastic power flow, (poster)

INFORMS Annual Meeting, (Anaheim, California, USA, Oct., 2021)
Data-driven piecewise linearization for distribution three-phase stochastic power flow, (presentation)

IEEE 4th Conference on Energy Internet and Energy System Integration (EI2), (Wuhan, Hubei Province, China, Oct., 2020)
Robust data-driven linearization for distribution three-phase power flow, (poster)

IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia), (Chengdu, Sichuan Province, China, May, 2019)
Asynchronous distributed dynamic economic dispatch inactive distribution networks, (presentation),
Distributed economic dispatch for active distribution networks with virtual power plants, (presentation),
Security evaluation under N-1 for active distribution networks coordinated with transmission grid, (poster)

IEEE 2th Conference on Energy Internet and Energy System Integration (EI2), (Beijing, China, Oct., 2018)
Risk assessment with high distributed generations penetration considering the interaction of transmission and distribution grids, (poster)

Courses, Skills and Teaching Experience

Courses: Advanced Power System Analysis, Electric Distribution Systems, Nonlinear Optimization, Optimization Methods for Power Systems, Power Systems and Market Operations, Introduction to Artificial Intelligence, Matrix Methods in Machine Learning, Energy Systems and Control, Distributed Control and Optimization of Power Systems

Teaching Assistant: *Power Systems and Market Operations*, Tsinghua University, Spring, 2020

Strengths: data-driven method, optimization, distribution system, electricity market, renewable energy resources

Programming: Matlab, Julia, Python, C, C++

Languages: English, Chinese